ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R does not explain why the image is inverted.
2	**(c)** -25 cm	f = -(far point) for objects at infinity.
3	**(b)** smaller wavelengths are scattered more by small particles	Rayleigh type scattering.
4	**(c)** A is true but R is false.	A is true; R is false: the image forms in front of the retina.
5	**(b)** -1 D and +1.5 D	Distance: f = -100 cm. Reading: 1/f = -1/40 + 1/25 = 3/200, P = +1.5 D.
6	**(a)** blue light more strongly than red light	Shorter wavelengths are scattered more.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Higher n means more bending.
8	**(b)** the ciliary muscles changing the shape of the lens	Accommodation by the ciliary muscles.
9	**(c)** larger than the wavelength of light	All colours scatter.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Shorter wavelength scatters more.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Red is scattered the least (1); The scattered light is dominated by shorter wavelengths (1).	1+1
12	Colours always separate (1); Because n differs for each colour (1).	1+1
13	Iris limits the entry of light (1); Lens changes its curvature to focus (1).	1+1
14	1/f = -1/50 + 1/25 = 1/50 (1); P = +2 D (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	1/f = -1/30 + 1/25 (1); 1/f = 1/150, f = 150 cm (1); P = +0.67 D (1).	1+1+1
16	Light reaches us after passing through a thick layer of air (1); Light is scattered many times by particles of all sizes (1); Colours mix and appear whitish (1).	1+1+1
17	At 10 cm (1); The lens must be thicker and have a shorter focal length (1); For 100 cm the lens is thinner (1).	1+1+1
	OR: Hypermetropia and myopia (1); Reading: 1/f = -1/100 + 1/25, P = +3 D (1); Distance: f = -2 m, P = -0.5 D (1).	1+1+1
18	Prism disperses the light (1); Edges appear coloured due to unequal deviations of colours (1); Violet and red are separated (1).	1+1+1
	OR: Sky is bright due to scattered sunlight (1); Stars' light is overpowered (1); At night no scattering (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) f = -90 cm, P = -1.11 D (1); (b) Concave lens (1); (c) f = -100 cm; image must be at 90 cm (1); 1/u = 1/v - 1/f = -1/90 + 1/100 = -1/900 (1); New far point is 900 cm (9 m), so correction is incomplete (1).	1+1+1+1+1
	OR: u = infinity, v = -2 m (1); f = -2 m (1); P = -0.5 D (1); Image at the far point, 2 m (1); It is virtual and erect for the lens (1).	1+1+1+1+1
20	(a) 1/f = -1/50 + 1/25 = 1/50 (1); P = +2 D (1); (b) f = 100 cm, 1/u = 1/v - 1/f = -1/50 - 1/100 (1); 1/u = -3/100 (1); u = -33.3 cm, so near point 33.3 cm (1).	1+1+1+1+1
	OR: (a) Red scatters least, so visible from far (1); (b) Yellow/red fog lamps penetrate better (1); (c) Earth sky is blue due to scattering (1); (c) Moon has no atmosphere, so a black sky (1); Scattering depends on wavelength and particle size (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** white light	The second prism reverses the dispersion.
2	**(c)** go undeviated	No relative refraction.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Atmospheric refraction.
4	**(c)** is nearly monochromatic	Single wavelength.
5	**(a)** violet travels slower than red	n(violet) > n(red).
6	**(a)** Both A and R are true and R is the correct explanation of A.	Averaging of variations.
7	**(a)** Pass a beam through water with a few drops of milk	Colloidal milk particles scatter light.
8	**(b)** Light of all colours gets scattered many times	Multiple scattering mixes colours.
9	**(c)** A is true but R is false.	A is true; R is false: red has the longest wavelength and is scattered least.
10	**(c)** is not deviated appreciably	Normal incidence along vertical layers.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distant rays are parallel and the eye can accommodate (1); Near objects need greater accommodation, which is lacking (1).	1+1
12	Hypermetropia (1); Near objects are focused behind the retina (1).	1+1
13	f = -60 cm (1); P = 1/(-0.6) = -1.67 D (1).	1+1
14	Their light passes through more atmosphere (1); Greater variation in the refraction (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Atmospheric particles are smaller than the wavelength (1); Short wavelengths scatter more, so noon sky is blue (1); Long path at sunset scatters away blue, leaving red (1).	1+1+1
16	Sun emits less violet (1); Upper atmosphere absorbs violet (1); Our eyes are less sensitive to violet (1).	1+1+1
17	Dark/black (1); Moon has no atmosphere (1); Without scattering the Sun's light isn't spread (1).	1+1+1
	OR: Red bends the least and returns at a larger angle (1); Violet bends the most and returns at a smaller angle (1); Hence red appears at the top (outer) and violet at the bottom (1).	1+1+1
18	Tiny droplets scatter the light (1); Scattering spreads light in all directions (1); Light seems to surround the source (1).	1+1+1
	OR: In the dark more light is needed (1); In bright light the pupil shrinks to protect the retina (1); Controlled by the iris muscles (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	-2 D means f = -50 cm (1); For u = infinity, v = f = -50 cm which is the far point (1); -1 D: f = -100 cm (1); 1/u = 1/v - 1/f = -1/50 + 1/100 (1); u = -100 cm (1).	1+1+1+1+1
	OR: Presbyopia: weak ciliary muscles (1); Upper lens: concave for distance (1); Upper f = -2 m (-0.5 D) (1); Lower lens: convex for reading (1); Lower f = 33.3 cm (+3 D) (1).	1+1+1+1+1
20	Refraction as light enters the drop (1); Dispersion since n varies with colour (1); Internal reflection at the back (1); Sun must be behind the observer, low in the sky (1); Red bends least and lies at the top (1).	1+1+1+1+1
	OR: In dim light the iris dilates the pupil (1); In bright light it contracts the pupil (1); For near objects the ciliary muscles contract (1); Lens thickens, focal length decreases (1); For distant objects the opposite (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option

attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** slightly less	No advance sunrise or delayed sunset.
2	**(a)** Both A and R are true and R is the correct explanation of A.	No scattering in space.
3	**(b)** it is much closer and extended	Averaging effect.
4	**(d)** the far point of the eye is 50 cm	f = -50 cm.
5	**(a)** Both A and R are true and R is the correct explanation of A.	Presbyopia with myopia.
6	**(d)** bifocal lenses	Two corrections are needed.
7	**(c)** about 67 cm	1/v = 1/40 - 1/25 = -3/200.
8	**(a)** Both A and R are true and R is the correct explanation of A.	The observer must stand between the Sun and the rain to receive the returned light.
9	**(d)** rod cells	Rods are light-sensitive.
10	**(d)** 2 m	f = -2 m.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	f = -1/3 m = -33.3 cm (1); Far point = 33.3 cm (1).	1+1
12	n is larger for violet (1); Violet travels slower in glass (1).	1+1
13	Light is monochromatic (1); Only one wavelength exists (1).	1+1
14	Slab faces are parallel (1); Prism faces are inclined, so the dispersion adds up (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Light refracts at the first surface and disperses (1); It reflects once inside at the back surface (1); It refracts again as it exits, with violet deviated the most (1).	1+1+1
16	Clear air has small molecules scattering mainly blue (1); Dust scatters all colours (1); So the dusty sky looks whitish (1).	1+1+1
17	Concave lens makes objects look smaller (1); Convex lens makes objects look bigger (1); Concave helps far vision (1).	1+1+1
	OR: v = 2 cm, u = -25 cm (1); 1/f = 1/2 + 1/25 = 27/50, f = 1.85 cm (1); Possible only if the lens can shorten its focal length to 1.85 cm (1).	1+1+1
18	The lens itself becomes cloudy (1); Spectacles cannot clear it (1); Surgery replaces the lens (1).	1+1+1
	OR: Rods work in dim light (1); Cones work in bright light and detect colour (1); Both convert light into electrical signals (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cornea and lens correspond to the lens system (1); Iris and pupil correspond to aperture and diaphragm (1); Retina corresponds to the film or sensor (1); Ciliary muscles change the focal length, whereas a camera changes the lens position (1); Optic nerve corresponds to the cable carrying data (1).	1+1+1+1+1
	OR: Take two glass tanks, one with pure water and one with water and a few drops of milk (1); Pass a torch beam through each (1); Milk particles are colloidal and scatter the light (1); The beam is visible in the milk water (1); Pure water has no particles to scatter light (1).	1+1+1+1+1
20	Longer path scatters blue away so red remains (1); Rays from the lower and upper edges are refracted differently (1); The disc looks flattened (1); Atmospheric refraction bends rays over the horizon (1); Sun is seen about 2 minutes after the actual sunset (1).	1+1+1+1+1

	OR: Myopia: f = -125 cm (1); P = -0.8 D, concave (1); Hypermetropia: 1/f = -1/100 + 1/25 = 3/100 (1); f = 33.3 cm, P = +3 D, convex (1); Concave diverges for myopia; convex converges for hypermetropia (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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